Session 1 - Paradigm Justification Statement

1. Research topic and context

The study concerns production stopes mined by sublevel stoping at one underground mine in southern Peru. The researcher reports high explosive consumption, oversize fragments, elevated transport costs relative to monthly budget references, and missed monthly tonnage targets; these observations require verification against records. A prototype will connect fragmentation prediction with economic assessment of design alternatives. Its immediate relevance concerns this operation, without claiming generalization across Peru.

2. Preliminary research question

To what extent can a fragmentation-prediction prototype support design decisions that improve the balance of drilling, explosives, loading, transport and secondary-breaking costs in this sublevel-stoping operation?

3. Selected paradigm and justification

Design science is proposed because the intended contribution is to build and evaluate a decision-support artifact. Diagnosis, requirements, development and evaluation will be connected to the current standard design. An exclusively positivist orientation is not selected as the main approach because the purpose extends beyond testing associations to developing an evaluated solution, although quantitative hypothesis testing will support evaluation. Interpretivism is not selected as the main orientation because meaning-making and lived experience are not the central research questions. Mixed methods are not selected because no necessary qualitative question has yet been identified. These approaches answer different purposes and are not inferior alternatives. Artifact utility, performance and limitations will determine the coherence of the approach [M1].

4. Implications

The student reports 120 historical blasts and projects approximately 180 new events over at least six months, subject to data quality and permissions. Simple and ML predictors will be compared without assuming a winner; development and prospective validation will be separated. Split Desktop measurements after transport describe material at that stage, not directly at the original face. The prototype and reproducible protocol are candidate contributions whose novelty requires literature review. Mining Technology and Mining are preliminary thematic publication venues, not guaranteed outlets [R01, R03].

5. One doubt or tension

General Management authorization for design trials is the main uncertainty; academic-use and publication permissions must also be clarified. If trials are not authorized, conclusions will concern predictions and analytical comparisons, without claiming field-demonstrated causal savings.

References: bibliography, M1, R01 and R03. Draft for the student's review and defense.